

EDA REPORT

Title: Fine Particulate Matter Exposure and Public Health Outcomes in Washington State

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Summary of research questions:

1. Is there a statistically significant correlation between county-level annual mean PM2.5 concentration and hospitalization rates for respiratory disease in Washington State in 2022?
This question investigates whether increased exposure to fine particulate matter is associated with higher respiratory hospitalization rates.
2. Which Washington counties have the highest PM2.5 concentrations, and do these correlate to counties with the worst respiratory health outcomes?
This question identifies geographic patterns in air quality and health burdens across Washington counties, and examines whether high-pollution counties consistently show worse health outcomes.
3. Can county-level PM2.5 concentrations predict respiratory hospitalization rates using linear regression?
This question evaluates the predictive power of PM2.5 concentrations on respiratory health outcomes using statistical modeling.

Motivation: As our climate changes, wildfires pose an increasingly frequent and dangerous risk, particularly in Washington State. Wildfire smoke raises severe public health risks due to high concentrations of fine particulate matter (PM2.5), gases, and toxins, which can cause respiratory issues, heart attacks, and long-term health risks. We choose to analyze the 2022 fire season because it had an extremely long and severe impact in Washington state. As students living in the Pacific Northwest, this issue is extremely relevant and directly affects our communities and the well-being of our state.

Datasets:

Dataset 1: EPA Air Quality System (AQS) Daily PM2.5 Data, 2022 - [Download Daily Data | US EPA](#)

This dataset contains daily PM2.5 measurements for Washington State in 2022. We will take daily values in the Daily Mean PM2.5 Concentration column into annual county-level averages and merge them with the CDC PLACES data on county names.

Dataset 2: CDC PLACES: Local Data for Better Health, County Data 2024 Release - [PLACES: Local Data for Better Health, County Data 2024 release | Data | Centers for Disease Control and Prevention](#)

This dataset provides model-based survey data, based on 2021-2022 survey data. We use the StateDesc column to filter Washington State counties and use the CASTHMA (current asthma prevalence among adults) column as our respiratory health outcome variable.

Note: CDC PLACES provides model-based estimates rather than direct measurements, so we pair it with 2022 EPA data to maintain temporal consistency with the 2021–2022 survey period. For this analysis, we used crude prevalence estimates from CDC data.

Method:

Research Question 1: Load the EPA AQS 2022 daily PM2.5 CSV filtered to Washington State, Group by the COUNTY column and compute the annual mean of the Daily Mean PM2.5 Concentration column for each county, Load the CDC PLACES 2024 County Data CSV and filter rows where State = "Washington", Filter to 2022, asthma measure, crude prevalence, Merge the two dataframes on the county name, Use SciPy library functions to compute the correlation coefficient and p-value between mean PM2.5 and asthma prevalence

Research Question 2: Using the merged dataframe from Q1, sort counties by mean PM2.5 concentration in descending order, Identify the top 10 counties by annual mean PM2.5 level, Compare their asthma prevalence values to the Washington State county average, Use SciPy library functions to test whether the top 10 highest PM2.5 counties have statistically significantly higher asthma prevalence than the remaining counties, reporting the t-statistic and p-value

Research Question 3: Using the merged dataframe from Q1, set annual mean PM2.5 as the independent variable (X) and asthma prevalence as the dependent variable (Y), Fit a linear regression model and report the R-squared value, regression coefficient for PM2.5, p-value, and 95% confidence interval, If the PM2.5 coefficient p-value is less than 0.05 and R-squared is meaningfully above 0, conclude that PM2.5 is a statistically significant predictor of respiratory health outcomes, Visualize the regression fit using a seaborn regression plot overlaid on the scatter plot from Q1

EDA-Specific Analysis: For the EDA, we will examine the structure and size of each dataset, identify and document any missing values, generate 7-number summaries for the quantitative variables, create at least 2 visualizations per dataset, and validate the assumptions for statistical testing.

EDA Results:**Dataset 1: EPA PM2.5 Data (2022)****Size of dataset:**

The EPA dataset contains 23,110 rows and 22 columns after cleaning. Each row represents a daily PM2.5 measurement recorded at a specific monitoring site in Washington State in 2022. The columns include pollution measurement variables, temporal information, and geographic identifiers.

Missing Data Analysis:

I identified missing values using the built-in pandas function `.isnull().sum()` and found some missing data for "local site name" and for "CBSA code" and "CBSA name". CBSA stands for "core based statistical area", which is a geographic region centered around an urban area, but there are no missing values for "County", which is the geographic variable we will be using for our analysis. The other variables of interest are "Daily Mean PM2.5 Concentration", "Daily AQI Value", and "Date". Because the missing does not affect the variables of interest, no additional imputation was necessary. Additionally, during the exploratory analysis I noticed a small number of negative PM2.5 concentration values (min = -2.4), which are not physically meaningful and likely caused by sensor malfunctions so they were removed during data cleaning.

Variables of Interest:

The primary variables of interest for this dataset are:

Daily mean PM2.5 Concentration- This represents the concentration of fine particulate matter in the air,

measured in $\mu\text{g}/\text{m}^3$. This is the main exposure variable for our research questions because of its prevalence in wildfire smoke.

Daily AQI Value- This represents the Air Quality Index, a standard measure of air quality based on multiple pollutant levels. This is an interpretable measure of pollution severity and most commonly used.

County- The categorical variable that we will use to identify a geographic region and aggregate PM2.5 exposure.

Date- This variable identifies the date of each measurement and is used to analyze temporal patterns and seasonal variation.

Summary Statistics:

SUMMARY STATISTICS (PM2.5): count - 23110.000000; mean - 7.042419; std - 9.319613; min - 0.00; 25% - 2.900000; 50% - 4.600000; 75% - 8.000000; max - 255.900000

Name: Daily Mean PM2.5 Concentration, dtype: float64

SUMMARY STATISTICS (AQI): count - 23110.000000; mean - 32.098875; std - 23.658155; min - 0.000000; 25% - 16.000000; 50% - 26.000000; 75% - 44.000000; max - 362.000000

Name: Daily AQI Value, dtype: float64

These statistics indicate that most PM2.5 values are relatively low, but the large maximum value and high standard deviation suggest the presence of extreme pollution events, likely associated with wildfire activity. The distribution is strongly right-skewed, meaning that most days have low pollution levels, with occasional spikes.

Dataset 2: CDC PLACES Health Data (2024)

Size of dataset:

The CDC PLACES dataset for Washington State contains 3,120 rows and 22 columns. After filtering to Washington State, 2022, asthma measure, and crude prevalence, we have 39 rows - one per Washington county.

Missing Data Analysis:

I identified missing values using the built-in pandas function `.isnull().sum()` and found that the `Data_Value_Footnote_Symbol` and `Data_Value_Footnote` columns have 100% missing values (3,120 missing each). These columns contain footnotes for data suppression flags and are not needed for our analysis. However, the columns of importance, `LocationName` (county), `Data_Value` (asthma prevalence), `Year`, `Measure`, and `Data_Value_Type`, have no missing values. The missing data does not affect our variables of interest.

Variables of Interest:

The primary variables of interest for this dataset are:

LocationName- This tells us the name of the County, and is used as the geographic identifier in this data.

Data_Value- This tells us the crude asthma prevalence percentage, which is our primary health outcome.

Year- identifies the year of data collection. We filter to 2022 to match the temporal window of our EPA PM2.5 data

Measure- indicates which health metric is being reported, I filtered specifically to asthma measures to isolate data on respiratory outcomes.

Data_Value_Type- indicates whether the prevalence estimate is crude or age-adjusted. For this project, I have selected only the crude prevalence for simplicity, although in Washington Counties, both measures were almost the same.

Summary Statistics (Asthma Prevalence - Crude):

Summary Statistics: CDC Asthma Prevalence:

Count: 39, Mean: 11.81, Std Dev: 0.64, Min: 9.30, Q1: 11.55, Median: 11.90, Q3: 12.20, Max: 12.90

These statistics summarize asthma prevalence across Washington's 39 counties. The low standard deviation indicates that asthma prevalence is relatively consistent across counties. The distribution is roughly symmetric, as the mean and median are very similar. Unlike Sophia's PM2.5 data, which showed extreme right-skew with occasional pollution spikes, the asthma prevalence data has no extreme outliers. This suggests that while air pollution varies dramatically across days and locations, the health outcome of asthma is more uniformly distributed across Washington's counties.

Merged Dataset Results

Size of merged dataset:

The merged dataset contains 31 rows and 3 columns after joining EPA PM2.5 annual means with CDC asthma prevalence on the county name. Each row represents a Washington county that had both PM2.5 monitoring data and CDC asthma survey data available for 2022.

Missing Data:

Eight Washington counties (Garfield, Columbia, Wahkiakum, San Juan, Skamania, Ferry, Pend Oreille, and Stevens) had no PM2.5 monitoring data in the EPA dataset. These are probably rural counties with little ability to measure PM2.5, so they are dropped from this study. Otherwise, the merged dataset has no missing values within the 31 matched counties.

Visualizations

Visualization 1: Distribution of Daily PM2.5 Levels in Washington State (2022)

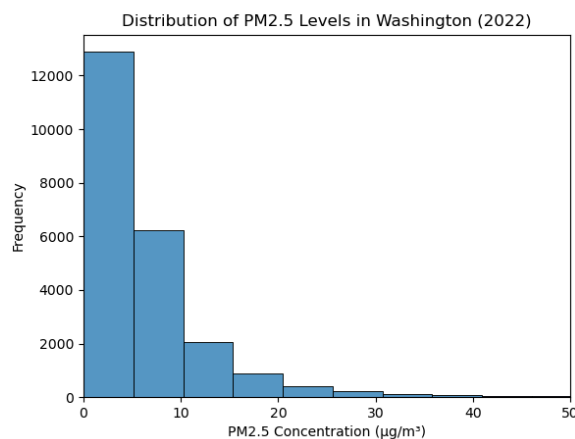


Figure 1 Caption:

Distribution of daily PM2.5 concentrations showing most values are low, with a right-skewed tail

representing extreme pollution events.

This histogram displays the distribution of daily PM2.5 concentrations across Washington State in 2022. A histogram was chosen to visualize how frequently different pollution levels occur. The plot shows that most PM2.5 values are concentrated at lower levels (below 10 $\mu\text{g}/\text{m}^3$), while a small number of extreme values extend far to the right. These high values indicate occasional significant pollution events, which are likely associated with wildfire smoke.

Visualization 2: Average Daily PM2.5 Levels in Washington State (2022)

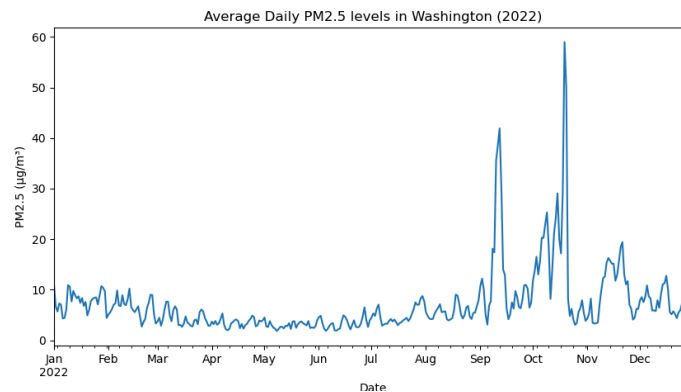


Figure 2 Caption:

Time series of PM2.5 levels showing seasonal spikes, particularly during late summer wildfire season.

This time series plot shows the average PM2.5 concentration across all monitoring sites over time. A line plot was chosen to analyze changes in pollution levels across the year. The plot reveals relatively low pollution levels during most of the year, with noticeable spikes occurring in late summer and early fall. These spikes are consistent with wildfire season in Washington State, suggesting that wildfires significantly impact air quality.

Visualization 3: Top 10 Counties by Average PM2.5 Levels (2022)

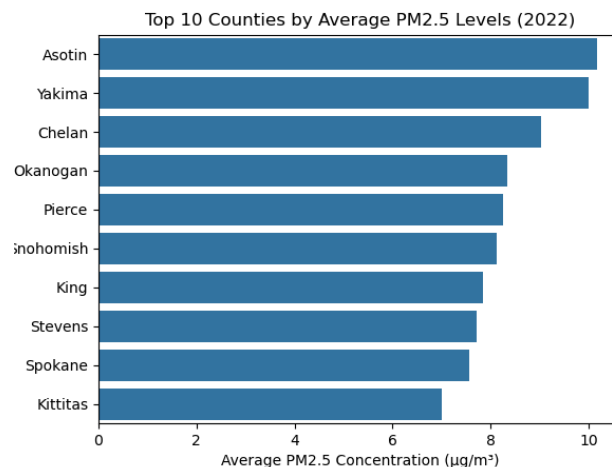


Figure 3 Caption:

Bar chart of counties with highest average PM2.5 levels, highlighting geographic variation in pollution exposure.

This bar chart compares average PM2.5 levels across counties. A bar chart was chosen because it clearly displays differences between categorical groups. The plot shows that certain counties experience consistently higher levels of pollution, indicating geographic disparities in air quality. These differences may be influenced by factors such as proximity to wildfire-prone regions or local environmental conditions.

Visualization 4: Relationship Between PM2.5 and AQI

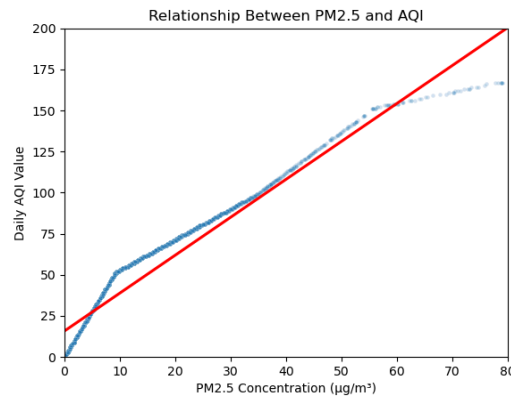


Figure 4 Caption:

Scatterplot showing a strong positive linear relationship (Correlation coefficient: 0.9089) between PM2.5 concentration and AQI where higher particle concentrations correspond to worse air quality.

This scatterplot illustrates the relationship between PM2.5 concentration and AQI. A scatterplot was chosen to visualize the relationship between two quantitative variables, and a linear regression line is included to summarize the overall trend. The plot shows a strong positive association, meaning that higher PM2.5 concentrations correspond to worse air quality. To improve visibility and avoid distortion from extreme outliers, the x-axis was limited to values below $80 \mu\text{g}/\text{m}^3$. This focuses on the range where the majority of observations fall, allowing for clearer interpretation of the general trend.

Visualization 5: Summary of Asthma Prevalence Across Washington Counties

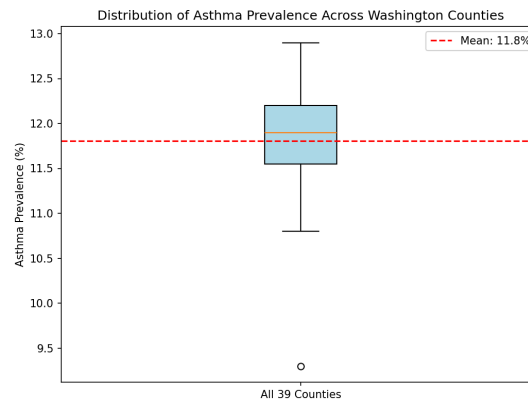


Figure 5 Caption:

Histogram showing the distribution of adult asthma prevalence across all 39 Washington counties. The distribution is roughly symmetric with a mean of 11.8%, suggesting asthma rates are relatively consistent across the state.

This box plot provides a concise statistical summary of asthma prevalence across Washington counties. A box plot was chosen because it efficiently displays the five-number summary (minimum, first quartile, median, third quartile, and maximum) and highlights any potential outliers. The box shows that the middle 50% of counties fall between 11.6% and 12.2%, with a median of 11.9%. The whiskers extend to the maximum and minimum values, and one outlier is present.

Visualization 6: Distribution of Asthma Prevalence Across Washington Counties

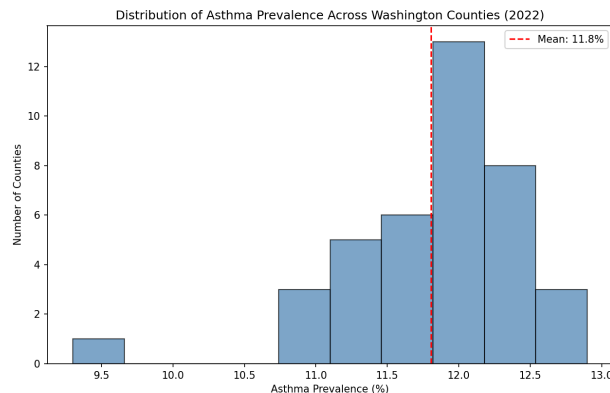


Figure 6 Caption:

Box plot summarizing asthma prevalence across Washington counties. The median is 11.9%, with the middle 50% of counties falling between 11.6% and 12.2%. One mild outlier is present (King County at 9.3%), but it falls within normal range and does not violate assumptions for testing.

This histogram displays the distribution of asthma prevalence percentages across Washington State counties in 2022. A histogram was chosen to visualize how frequently different asthma rates occur across counties. The plot shows that most counties have asthma prevalence between 11.5% and 12.5%, with a

roughly symmetric distribution around the mean of 11.8%, other than one county around 9.3%. No points appear beyond 1.5 times the interquartile range, confirming no extreme outliers

Visualization 7: Distribution of Asthma Prevalence Across Washington Counties

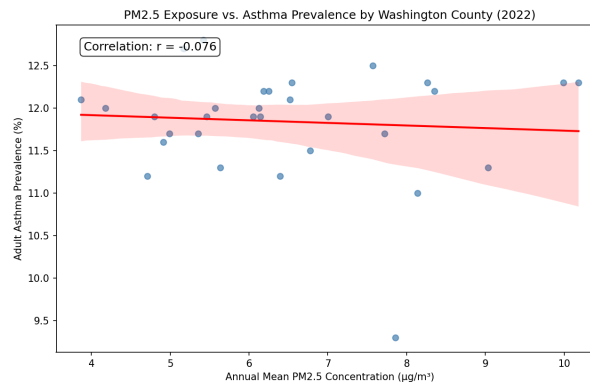


Figure 7 Caption:

Scatter plot showing the relationship between annual mean PM2.5 concentration and asthma prevalence across 31 Washington counties. The correlation coefficient is $r = -0.076$, indicating no significant linear relationship between PM2.5 exposure and asthma prevalence in 2022.

This scatter plot illustrates the relationship between PM2.5 exposure and asthma prevalence across Washington counties. A scatter plot was chosen to visualize the association between two quantitative variables. Each point represents one county that had both PM2.5 monitoring data and CDC asthma data. Notice that the correlation is almost 0 and that the line of best fit is almost horizontal; there seems to be very little linear relationship.

Result Validity: Null Hypotheses and Assumption Checking

Null Hypotheses for Statistical Testing:

RQ1 $H_0: \rho = 0$ (no correlation between PM2.5 and asthma prevalence)

RQ2 $H_0: \mu_{\text{high}} = \mu_{\text{other}}$ (top 10 PM2.5 counties have same asthma rate as other counties)

RQ3 $H_0: \beta = 0$ (PM2.5 does not predict asthma prevalence)

Assumption Verification for Pearson Correlation and Linear Regression:

Validating assumptions for Pearson correlation and linear regression ensures accurate results through ensuring that data is continuous, linear, no severe outliers, and a normal distribution. We will do this through creating the following: Scatterplots for linearity - figure 7, Residual plots for homoscedasticity, Normality - histogram looks roughly normal, and Independence - counties are geographically distinct

Challenge Goals:

Result Validity: For each research question, we will verify our findings using statistical significance testing. Specifically, we will use the SciPy and Pandas libraries to measure the strength and direction of relationships between variables. We will report p-values (significant if $p < 0.05$) and confidence intervals, and interpret what they mean in context. This statistical inference goes beyond what was taught in class.

New Library: We will use statsmodels and SciPy, which are Python libraries not covered in CSE 163, to perform linear regression analysis and technical computing, examining how smoke exposure predicts health outcomes over time.

Work Plan:

Task 1 - Download, clean, and merge datasets (estimated 6 hours): Download EPA 2022 and CDC Places 2024 CSV's. Then we need to filter to Washington state, aggregate PM 2.5 to county-level annual means, and merge datasets by the name of the county. This will be done by both Melodie and Sophia together

Task 2 - Exploratory Data Analysis - (estimated 2 hours): Calculate summary statistics, check for missing counties or null values, create initial visualizations. This will be done by both together

Task 3 - Research Question 1 Analysis (estimated 2 hours): Correlation analysis using SciPy, significance testing, and scatter plot with regression line, Will be done by Sophia

Task 4 - Research Question 2 Analysis (estimated 2 hours): County ranking by PM2.5, group comparison using SciPy, and the bar chart, Done by Melodie

Task 5 - Research Question 3 Analysis (estimated 2 hours): Linear Regression using statsmodels, interpretation, and regression visualization, This will be done by both Melodie and Sophia together

Task 6 - Report Writing (estimated 3 hours): Write up findings, interpretations, and create final visualizations. This will be done by both Melodie and Sophia together

Workflow: Sophia and Melodie will work together on certain tasks as well as independent research. We will use GitHub to share and version control our code, committing after each task so either partner can pick up where the other left off. We will meet regularly to work on the collective portions, as well as share our independent work and provide support.

Testing: We tested our EDA code using assert statements to verify that the filtered data contained only Washington State. We also used manual verification in comparing merged county names to ensure proper matching. Furthermore, we visually inspected all visualizations for obvious errors. Our results are trustworthy: All data cleaning is programmatic and reproducible. We verified that no data was lost during merging. Summary statistics were cross-checked using both pandas .describe() and manual calculations on small subsets. Our testing suite covers data integrity, boundary conditions, and merge correctness. All code passes flake8 and follows CSE 163 code quality standards

Collaboration- All code passes flake8

Melodie has familiarity with CDC datasets, but still consulted the CDC PLACES documentation to understand the long data format and how to properly filter for asthma prevalence measures, particularly with the formatting for crude prevalence and age-adjusted prevalence. Stack Overflow was used for pandas merge and groupby syntax questions. No other sources beyond TA and professors were consulted outside of our group.